MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 15

1. *Find the eigenvectors and eigenvalues of the following matrices:*

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \begin{bmatrix} 4 & 2 & 2 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}.$$

(Challenge question)*: For each of the above matrices, are the eigenvectors corresponding to different eigenvalues orthogonal?*

We begin with computing the characteristic polynomial:

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -1 & 2 \\ 1 & 1-\lambda & 2 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)\begin{vmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{vmatrix}$$

$$= -(\lambda-2)((1-\lambda)(1-\lambda)+1) = -(\lambda-2)(\lambda^2-2\lambda+2) = -(\lambda-2)(\lambda-(1+i))(\lambda-(1-i))$$

Next we find the eigenvalues of A, which are the zeros of $p(\lambda)$. Eigenvalues in this case are $\lambda = 2, 1+i, 1-i$. Now we find the eigenvectors for each eigenvalue (one eigenvalue at a time):

($\lambda = 2$) We will find a basis of $\text{Nul}(A - 2I)$:

$$A - 2I = \begin{bmatrix} -1 & -1 & 2 \\ 1 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to -R_1} \begin{bmatrix} 1 & 1 & -2 \\ 1 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 1 & -2 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 \to -\frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ is an eigenvector for A corresponding to $\lambda = 2$ and span $\left\{ \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \right\}$ is the eigenspace of A corresponding to $\lambda = 2$.

($\lambda = 1+i$) Now we will find a basis for $\text{Nul}(A - (1+i)I)$:

$$A - (1+i)I = \begin{bmatrix} 1-(1+i) & -1 & 2 \\ 1 & 1-(1+i) & 2 \\ 0 & 0 & 2-(1+i) \end{bmatrix} = \begin{bmatrix} -i & -1 & 2 \\ 1 & -i & 2 \\ 0 & 0 & -1-i \end{bmatrix}$$

$$\xrightarrow{R_1 \to iR_1, R_3 \to \frac{1}{-1-i}R_3} \begin{bmatrix} 1 & -i & 2i \\ 1 & -i & 2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & -i & 2i \\ 0 & 0 & 2-2i \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2 \to \frac{1}{2-2i}R_2, R_3 \to R_3 - R_2, R_1 \to R_1 - 2iR_2} \begin{bmatrix} 1 & -i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $\begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = 1+i$ and span $\left\{ \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix} \right\}$ is the eigenspace of A corresponding to $\lambda = 1 + i$.

($\lambda = 1 - i$) For $\lambda = 1 - i$, we *could* do a calculation similar to what we did for $\lambda = 1 + i$ above. However, since A is a matrix with real entries, we can just use the fact that since $1 - i$ is the complex conjugate of $1 + i$, then the complex conjugate of the eigenvector for $\lambda = 1 + i$ above will be an eigenvector for $\lambda = 1 - i$. Thus $\begin{bmatrix} -i \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = 1 - i$ and span $\left\{ \begin{bmatrix} -i \\ 1 \\ 0 \end{bmatrix} \right\}$ is the eigenspace of A corresponding to $\lambda = 1 - i$.

For the second matrix, $\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 & 2 \\ -2 & -\lambda & 0 \\ -2 & 0 & -\lambda \end{vmatrix}$.

Using cofactor expansion along the second column, we have

$$\det(A - \lambda I) = -2 \begin{vmatrix} -2 & 0 \\ -2 & -\lambda \end{vmatrix} + \lambda \begin{vmatrix} 4-\lambda & 2 \\ -2 & -\lambda \end{vmatrix}$$

$= 4\lambda + \lambda(\lambda^2 - 4\lambda + 4) = \lambda(\lambda^2 - 4\lambda + 8)$.

Applying the quadratic formula to $\lambda^2 - 4\lambda + 8$, we have

$$\det(A - \lambda I) = \lambda(\lambda - (2 + 2i))(\lambda - (2 - 2i)).$$

Therefore the eigenvalues are $\lambda = 0, 2 + 2i, 2 - 2i$.

For $\lambda = 0$:

$$A - \lambda I = \begin{bmatrix} 4 & 2 & 2 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{4}R_2, R_3 \to R_3 - R_2} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - \frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

so the eigenspace corresponding to $\lambda = 0$ is span$\{ \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \}$.

Next, for $\lambda = 2 + 2i$:

$$A - \lambda I = \begin{bmatrix} 2-2i & 2 & 2 \\ -2 & -2-2i & 0 \\ -2 & 0 & -2-2i \end{bmatrix} \xrightarrow{R_2 \to R_2 + \frac{2}{2-2i}R_1, R_3 \to R_3 + \frac{2}{2-2i}R_1} \begin{bmatrix} 2-2i & 2 & 2 \\ 0 & -1-i & 1+i \\ 0 & 1+i & -1-i \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 2-2i & 2 & 2 \\ 0 & -1-i & 1+i \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2-2i}R_1} \begin{bmatrix} 1 & \frac{4+4i}{8} & \frac{4+4i}{8} \\ 0 & -1-i & 1+i \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 + \frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & 1+i \\ 0 & -1-i & 1+i \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 \to \frac{1}{-1-i}R_2} \begin{bmatrix} 1 & 0 & 1+i \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

So the eigenspace of $\lambda = 2 + 2i$ is $\text{span}\{\begin{bmatrix} -1-i \\ 1 \\ 0 \end{bmatrix}\}$. Taking the complex conjugate, the eigenspace of $\lambda = 2 - 2i$ is $\text{span}\{\begin{bmatrix} -1+i \\ 1 \\ 0 \end{bmatrix}\}$.

2. *Suppose A is an $n \times n$ matrix with complex coefficients (i.e., coefficients in $\mathbb{C}$). If $A^2 = -I$, what are the possible eigenvalues of A? If A has real coefficients (i.e., coefficients in $\mathbb{R}$), show that n* ***must*** *be even. Give an example of a matrix A such that $A^2 = -I$ and A has real coefficients.*

Suppose λ is an eigenvalue of A with eigenvector $\mathbf{v} \neq \mathbf{0}$. Then $A\mathbf{v} = \lambda\mathbf{v}$. Multiplying again by A on the left gives $A^2\mathbf{v} = A(\lambda\mathbf{v}) = \lambda A\mathbf{v} = \lambda \cdot \lambda\mathbf{v} = \lambda^2\mathbf{v}$. This is also equal to $-I\mathbf{v} = -\mathbf{v}$ (since $A^2 = -I$). Since $\mathbf{v} \neq \mathbf{0}$, we conclude that $\lambda^2 = -1$. Therefore the possible eigenvalues for A are $\lambda = \pm i$ (since these are the only complex numbers which satisfy the relation $\lambda^2 = -1$).

Next assume that the entries of the matrix A are real numbers. Then $\det(A)$ **must** be a real number. However, $\det(A)$ is also the product of the eigenvalues of A (taking into account multiplicity). There are n eigenvalues and they are all equal to $\pm i$. Therefore, if n is odd, then the determinant will be $\pm i$, whereas if n is even then the determinant will be ± 1. For example, if $n = 5$ and the eigenvalues are $i, i, i, -i, -i$, then the determinant will be $i^3(-i)^2 = i(-1)(-1) = i$ (which is imaginary). If $n = 6$ and the eigenvalues are $i, i, i, -i, -i, -i$, then the determinant will be $i^3(-i)^3 = i(-1)(-i)(-1) = 1$ (which is real).

If $n = 2$, we can take as an example

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

which clearly has real coefficients.